

# GAIADA Creative

Image and video generation and editing, Magnific-replacement upscaling, and a cross-department asset library — one production-and-supply subsystem powered by our own or rented GPU compute through a Creative Render Gateway, never a per-seat SaaS subscription.

| DOCUMENT                      | MODULE KEY | DATE       | OWNER                       | STATUS                                         | CLASSIFICATION |
|-------------------------------|------------|------------|-----------------------------|------------------------------------------------|----------------|
| Engineering<br>Blueprint v1.0 | creative   | 2026-07-23 | Creative Dept ·<br>Platform | EXPANSION<br>PLANNED ·<br>STUDIO<br>PROTOTYPED | Internal       |

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## 500 Executive Summary

What we are building, why the render gateway exists as a second deployable, and the shape of the four locked decisions this design conforms to.

Creative becomes a **platform-nest module vertical** ( `ModuleContract` key `creative`, tables `creative_*` — extending, not duplicating, the shipped `creative_assets` table from migrations 0031/0032) plus **one new deployable**, the **Creative Render Gateway** ( `render-gateway-go/`, the mirror of `ai-gateway-go` for diffusion workloads), plus the expanded Creatives department console. Six moves shape the architecture:

### 1 · THE ONLY DOOR TO GPU COMPUTE

A typed render job is created as an RLS-scoped `creative_render_jobs` row, dispatched to the gateway, which routes it to the cheapest capable **RenderBackend** — serverless GPU, self-hosted headless ComfyUI, or commercial API — filtered by capability + license class + cost cap + scope + health. Platform-nest owns job state; the gateway owns GPU keys and holds no tenant data at rest.

### 2 · LICENSE DISCIPLINE IS STRUCTURAL

Every ComfyUI workflow is a versioned JSON artifact carrying a model manifest with license classes. The gateway stamps provenance; platform-nest computes an asset's `license_class` as the weakest link of its derivation chain; the reuse gate refuses any chain that isn't fully commercial/licensed. FLUX quality is reachable only through the premium tier. v1 ships zero non-commercial workflows enabled.

### 3 · DAM = THE EXTENDED ASSET DOMAIN

Version stack, collections + brand kits, rights/licensing with expiry, provenance, CLIP embeddings → pgvector search + dedup, BLIP captions/auto-tags, and imgproxy as a stateless signed-URL rendition sidecar. Cross-dept consumers see approved assets only, through the module's BFF surface.

### 4 · TWO GATES, NOT ONE

The **spend gate** (per-client scope → ledger stop-loss chain client → tenant → global, fail-closed, WS4-suspended for automation and above-threshold/premium jobs) and the **reuse gate** (human WS4 approval promoting an asset to cross-dept/deliverable status, where the license chain is verified). Generative spend and generative publication are different risks.

### 5 · AI ASSIST IS LOCAL-FIRST

Prompt expansion, tag normalization, and alt-text ride `ai-gateway-go` (Hermes default, Claude polish flag). Vision models run as `analyze` render jobs — CLIP + BLIP on a cheap CPU worker backend — deliberately **not** the gateway `/embed`, which is not CLIP-space. Retrieval-shaped RAG stays in WS8.

### 6 · THE MAGNIFIC BLEED STOPS IN WEEK ONE

Phase 0 is a 1-day spike onto clarity-upscaler via Replicate (~\$0.019/run, ~10× cheaper than Magnific credits) before any platform code lands. Build order: **P0 contracts** → **P1 upscale (Magnific kill)** → **P2 image generate/edit** → **P3 DAM search/tagging + cross-dept supply** → **P4 video**. 27 tickets (CR-00–CR-26), /army-ready (\$12).

### READING THIS DOCUMENT

This is a **design blueprint**. The **Image Studio** persist surface it absorbs (migrations 0031/0032 + `CreativeController` + the platform-ui studio) is already **PROTOTYPED** — everything else in this document does not exist as code yet. It targets a future `creative` module registered as `0.0.0 · PLANNED` in `docs/modules/MODULES.md` on approval. Primary input: `creative-design.md` (v1.0, the architecture) and its companion `creative-foundation.md` (the research + the 4 locked decisions + the GPU-hosting reality, folded into §01/§06 below).

THE DEFINING HAZARDS

Where search-marketing is a *data + judgment* subsystem and SMM is a *judgment + publication* subsystem, Creative is the **production + supply** subsystem: its hazards are **GPU money burning by the second** and a **license boundary** — a non-commercial model's pixels must be structurally unable to reach a client deliverable.

§01 Scope & Service Lines

All five service lines are v1, phased — what ships, what's deliberately deferred, and the Magnific economics that justify building this at all.

| SERVICE LINE                                                  | V1 DELIVERS                                                                                                                                                                                                                     | DEFERRED                                                                                                                              |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Image enhancement / upscale</b> (the Magnific replacement) | Creative upscale (clarity-upscaler recipe: SDXL + ControlNet-Tile), precision upscale (Real-ESRGAN/DiffBIR), relight (IC-Light V1), batch jobs, before/after compare, preset tuning to Magnific parity                          | SUPIR (non-commercial — internal experiments only, ships disabled); Topaz-class video masters                                         |
| <b>Image generation</b>                                       | Text→image on the commercial-clean set (Qwen-Image / SDXL / Z-Image-Turbo), prompt-expansion assist, style presets per brand kit, FLUX hero tier behind the premium gate                                                        | LoRA training on client brand looks; OmniGen 2 unified gen+edit (prototype later)                                                     |
| <b>Image editing</b>                                          | Instruction editing (Qwen-Image-Edit), background removal (BiRefNet — not RMBG), inpaint/outpaint (ControlNet + IP-Adapter), per-network crop/format via renditions                                                             | Full layered canvas editor (v1 = form-canvas + mask-lite; InvokeAI-embed vs own-canvas is OQ-6)                                       |
| <b>Video generation + I2V</b>                                 | Wan 2.2 T2V / I2V / FLF2V (first+last-frame morph) on serverless GPU; RIFE interpolation + Real-ESRGAN video upscale post-pipeline; commercial-API hero shots (Veo synced audio, Kling/Runway) inside the \$100–300/mo envelope | Generative video editing/NLE (no OSS equivalent — assemble stays ffmpeg/Kdenlive, human-finished); object removal in video (immature) |
| <b>Asset management (DAM)</b>                                 | The shared library: versions, collections, brand kits, rights/licensing + expiry, provenance, CLIP/BLIP search + dedup, imgproxy renditions, approve-for-reuse, Shared Drive mirror                                             | Client-facing brand portals; smart-crop face detection (thumbor — add only if needed); federation to external DAMs                    |

Non-goals (v1)

- **No adopted external UIs** — ComfyUI and InvokeAI UIs are never exposed; the Creatives console is the only operator surface. ComfyUI runs headless at arm's length (GPL never linked, HTTP only).
- **No owned GPU hardware** — locked. The rented/serverless posture is the design; the owned-box tripwire is OQ-2.
- **No direct vendor AI calls** — LLM assist through ai-gateway-go only; GPU calls through the Render Gateway only.
- **No non-commercial model output on any client-visible path** — structural rule (§05/§07), not policy. SUPIR / IC-Light V2 / RMBG / SVD / FLUX-dev-free-weights are quarantined; HunyuanVideo additionally geo-caveated (EU/UK/S.Korea exclusion) and stays out of v1.
- **No payroll/billing changes** — GPU costs are metered in the creative ledger; invoicing them is the billing module's concern.
- **No new PM system** — creative work items ride the existing PM vertical; the module adds production/library domain only.

Fit with prior decisions — and two flagged conflicts

- Conforms to the **ERP holding-OS vision**: enablement via `enabled_modules` OR active `service_assignment`, WSD-3 module-sliced third-wall RLS — the Creative department in the agency company serves sibling companies without data bleed.
- Lands on the **dept-interface-template**: Creatives already has the two-level console (craft group "Studio": Image Studio · Asset Library). This design grows it to two craft groups (§08) and keeps the toolkit rule: no tab registered before its route exists.
- Fills the **mcp-hub reserved seam** — CLAUDE.md lists "Magnific `image.enhance` (no Gateway capability yet)" as a deferral; `creative.enhanceImage` (§07) closes it, backed by the Render Gateway rather than a Magnific subscription.
- Reuses, not duplicates: `clients`, `projects`, `files` storage backend, the `file` Cerbos policy (already how `creative_assets` authorizes), WS4 `automation_approvals` (origin widened), the SEO-proven ledger/stop-loss + one-shot-approval patterns.

FLAGGED CONFLICT 1 — SMM D-9

`smm-design.md` books generative-image credits in `social_usage_ledger`. Once this module exists, the render cost books **exactly once**, in `creative_usage_ledger`, attributed `requester_module='social'` + the requesting client; SMM's `tool_scope.ai.imageGen` toggle still gates the request at its own choke point, but its ledger must not double-book the GPU dollars. Noted for the owner as D-7 (\$14) rather than silently overriding SMM's decision log.

FLAGGED CONFLICT 2 — AGENCY CREATIVE-ASSET REVIEW (0006)

The agency vertical already has campaign-bound creative-asset review. That surface is **campaign deliverable review** and stays; the DAM is the production/library layer underneath. v1 keeps them separate; a bridging seam is a noted v2 item, not built now.

DESIGN CHANGE TO SHIPPED CODE (DELIBERATE, COORDINATED)

0031 gave `creative_assets` a plain tenant wall with an explicit "not a gated module" comment. This design **retrofits the third wall** and moves the controller into the module (D-3, \$04) — the 0031 comment is superseded by the module vertical materializing. The shipped HTTP contract is preserved.

The Magnific head-to-head (quality · quantity · price)

Apples-to-apples on Magnific's home turf — creative upscale/enhance (our stack also does gen/edit/video/DAM, not credited here):

- **Quality — parity for client work.** clarity-upscaler is the open engine powering ClarityAI.co (same SDXL + ControlNet-Tile recipe, same "hallucinated detail" glow). SUPIR would exceed Magnific but is non-commercial → off-limits, so on the clean stack we **match, not beat**. Magnific keeps a "turnkey polish" edge until we tune presets.
- **Quantity — uncapped vs. credit-capped.** Magnific: Pro ~500/mo, Premium ~1,500, Business ~5,000 credits. Ours: limited only by GPU budget/throughput. The same \$39 buys ~500 Magnific upscales vs. ~2,000 on Replicate; on an owned box it's unlimited at ~\$0 marginal.
- **Price + the seat point — the whole game.** Magnific bills **per account/seat with a credit cap**. Our stack is a backend service the whole team hits via the platform (MCP-mediated) — cost is **decoupled from headcount**. Zero seats, one backend; billing is compute, not people.

| TEAM OF 5, ~2,000 UPSCALES/MO                             | MONTHLY                                | SCALES WITH TEAM SIZE?            |
|-----------------------------------------------------------|----------------------------------------|-----------------------------------|
| Magnific — shared 1 Business acct                         | ~\$299 (concurrency-limited, ToS-gray) | —                                 |
| Magnific — proper per-seat (5× Premium)                   | ~\$495                                 | <b>Yes — +\$99 per new person</b> |
| <b>Ours — serverless</b> (Replicate/fal — chosen posture) | <b>~\$38 compute, unlimited seats</b>  | <b>No — flat</b>                  |
| Ours — owned/rented 24 GB box                             | ~\$248 flat, unlimited volume + seats  | No                                |

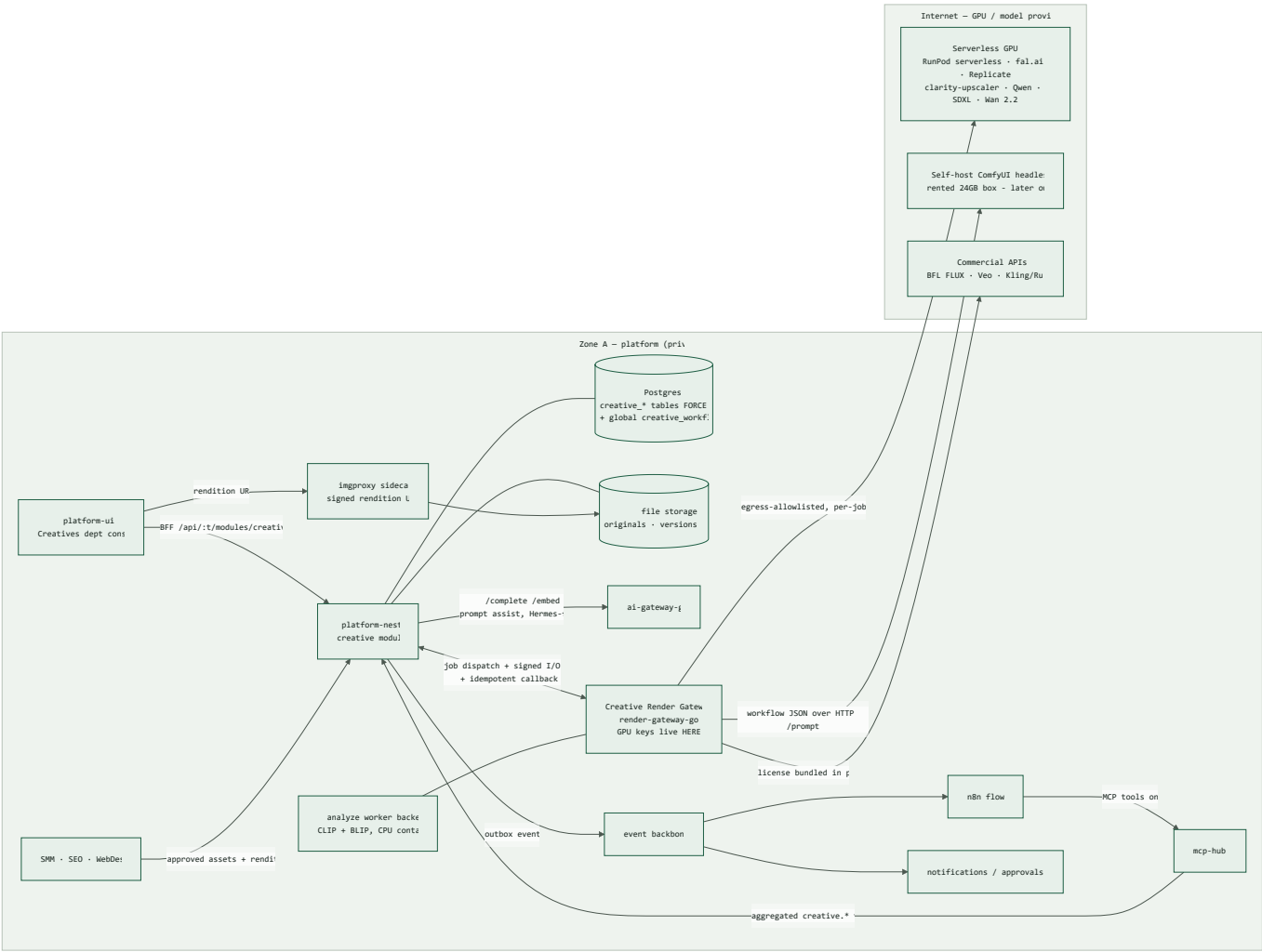
Unit cost: **~\$0.05–0.20/image** (Magnific credits) vs. **~\$0.019** (Replicate) vs. **~\$0 marginal** (owned box) — 3–10× cheaper per image, and the gap widens with every added person or batch. The win isn't quality (that's parity) — it's **cost decoupled from headcount + volume**, delivered through MCP.

HONEST CAVEATS

(1) Magnific is zero-effort; ours needs the Render Gateway built + light GPU ops. (2) This week clarity-upscaler on Replicate already saves ~10×/image, but the polished "zero-seat via MCP" state arrives only once the Gateway + console ship (P1). (3) Preset tuning is the work that buys parity. Break-even (even counting build) arrives fast past ~2–3 people or a few hundred images/mo — and unlike Magnific, the bill stops growing with the team.

§02 System Overview

All state, tenancy, judgment, and money controls live in Zone A. The Render Gateway is Zone-A infrastructure with egress-only GPU traffic; no inbound internet path exists to any of it.



**FIG. 02-1** – Master system diagram. Bytes flow storage → gateway → backend → gateway → storage via short-lived signed URLs platform-nest issues per job; the gateway persists nothing tenant-owned. imgproxy is a stateless sidecar reading the same storage, serving signed, on-the-fly renditions to the UI and to consuming departments.

Reading the diagram

- All state, tenancy, judgment, and money controls live in **Zone A**. The Render Gateway is Zone A infrastructure with **egress-only** GPU traffic (allowlisted hosts, per-job audit of every byte of client pixels leaving — \$03).
- Bytes flow storage → gateway → backend → gateway → storage via **short-lived signed URLs** platform-nest issues per job; the gateway persists nothing tenant-owned.
- **imgproxy** is a stateless sidecar reading the same storage, serving signed, on-the-fly renditions to the UI and to consuming departments.
- No inbound internet path exists to any of it; commercial-API webhooks are not used — **the gateway polls providers**.

§03 Trust Zones & Network

The WebDesk zone doctrine adapted to this subsystem's actual exposure — Creative has no public-facing surface at all; its risk is egress (client pixels to GPU clouds) and money.

| SURFACE                                 | ZONE | RULES                                                                                         |
|-----------------------------------------|------|-----------------------------------------------------------------------------------------------|
| creative module + DAM + ledger + scopes | A    | Standard FORCE-RLS + third wall; all writes through the module controller; approvals via WS4. |

| SURFACE                                                              | ZONE                          | RULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Creative Render Gateway</b><br>( <code>render-gateway-go</code> ) | <b>A, egress-only</b>         | Holds <b>all</b> GPU/model provider keys (RunPod, fal, Replicate, BFL, Google/Veo, Kling/Runway) — env now, OpenBao target-state; platform-nest never sees them. No inbound ports beyond the platform-nest-facing API (service-token auth; mTLS via the syncacct CA when enrolled). <b>Egress allowlist</b> enforced at DialContext exactly like ai-gateway-go — only registered backend hosts are reachable; every dispatch writes an egress-audit line <code>{job, tenant-hash, client-hash, backend, bytes_out, model}</code> . |
| Self-host ComfyUI target                                             | <b>external execution env</b> | A rented box (RunPod/Vast) reachable only from the gateway over its HTTP API ( <code>/prompt</code> , <code>/history</code> , ws progress); it holds <b>zero credentials into Zone A</b> . Treated as untrusted compute: outputs are content-type-validated and re-encoded on ingest (a compromised worker can corrupt a render, never reach the platform). GPL isolation is the same arm's-length HTTP pattern.                                                                                                                   |
| Serverless / commercial API backends                                 | <b>internet</b>               | Server-side keys only, per-call cost accounting, no client identifiers in prompts/filenames (opaque job ids); providers see pixels + prompt, nothing else. Clients who cannot accept cloud-GPU egress get <code>allow_cloud_gpu:false</code> in scope (OQ-5) — their jobs route only to the self-host target or refuse.                                                                                                                                                                                                            |
| imgproxy sidecar                                                     | <b>A</b>                      | Stateless MIT Go/libvips container; <b>signed URLs only</b> ( <code>IMGPROXY_KEY/SALT</code> , expiring), source restricted to our storage; never exposed to the public internet in v1 (client delivery goes through Drive/files, not hotlinks).                                                                                                                                                                                                                                                                                   |
| Input/output byte transfer                                           | <b>A</b> <code>gateway</code> | Platform-nest issues <b>short-lived signed GET (inputs) and PUT (outputs) URLs</b> per job; the gateway and backends never get storage credentials; callback references stored keys.                                                                                                                                                                                                                                                                                                                                               |
| Callbacks<br>(gateway → platform-nest)                               | <b>A</b>                      | <code>POST /api/internal/creative/render-callback</code> , idempotent by <code>job_id</code> , bearer service token (+ mTLS option), schema-validated; a forged callback without the per-job callback token is refused, and even a replayed valid one is idempotent (terminal-state transitions are one-way).                                                                                                                                                                                                                      |

## §04 Domain Model & Schema

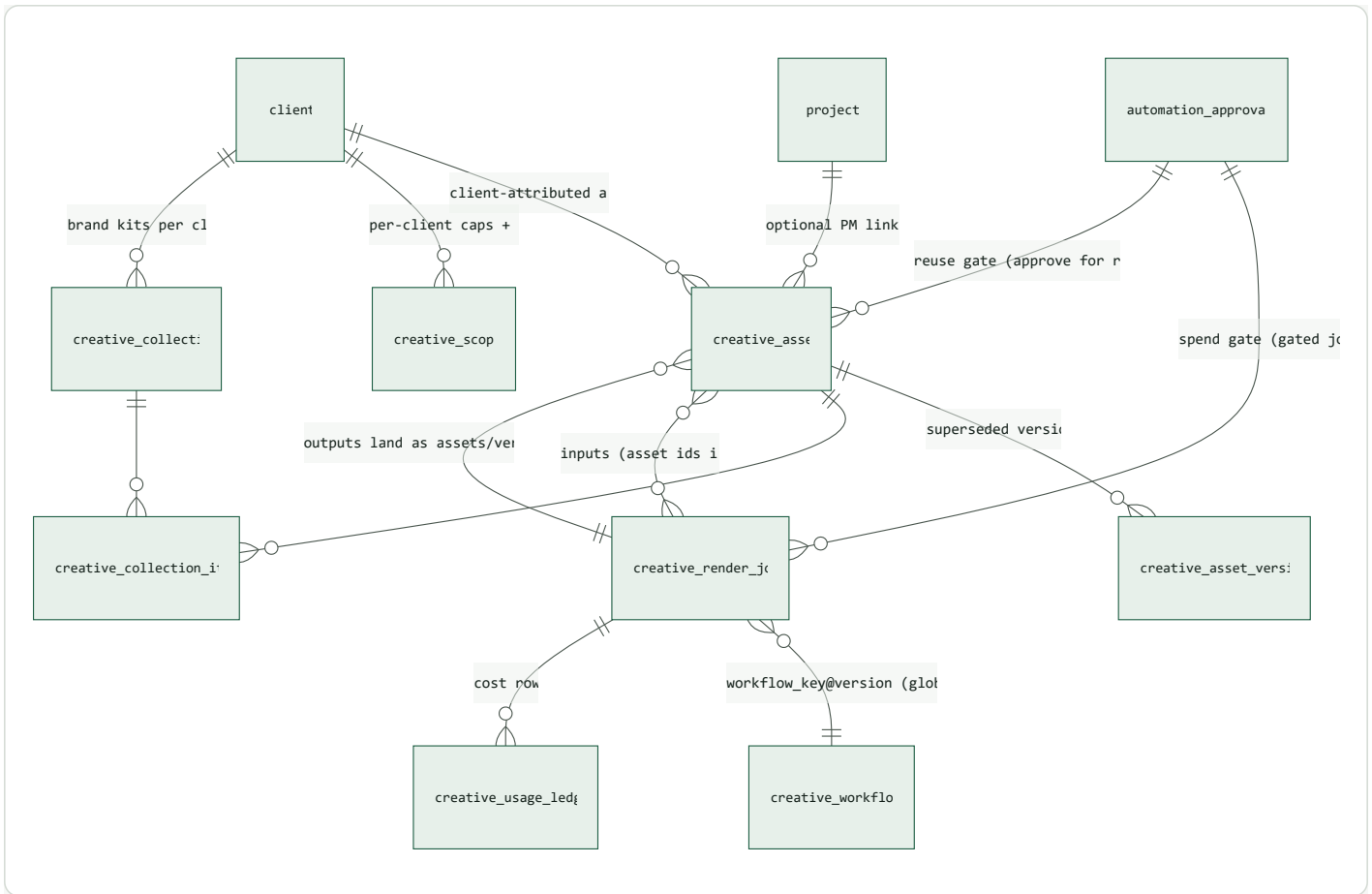
Every tenant table carries the third-wall RLS predicate; the render job is the state-machine spine; the extended asset row is the DAM.

### Design rules (inherited, not optional)

- Every tenant table: `tenant_id uuid NOT NULL REFERENCES companies(id)` , `origin_site` , `created_at/updated_at` , soft-delete `deleted_at` where user-facing.
- FORCE-RLS with the WSD-3 third wall** — policy predicate `tenant_id = ANY(app_current_tenants()) AND app_module_allowed('creative')` on both USING and WITH CHECK, byte-identical in shape to `0028_module_hr.sql` . App side reaches these tables only via `withTenants(tenants, { modules: ['creative'] })` .
- Migration numbering — stated assumption:** the module block starts at `0036_module_creative.sql` (+ `0037` if a follow-up is needed). Basis: as of 2026-07-23 the ledger runs through `0035_integration_connections_search_providers.sql` (the SEO design's "0034+" was drawn down by the search module). CR-01 takes the **next unused number at merge time** and coordinates with SMM-01 (which also targets next-unused) — whichever merges second rebases. No in-migration GRANTS; no `sync_app` grants (creative tables do not sync in v1).
- Money: **GPU/provider cost is `numeric(12,6)` USD** (serverless unit prices go to fractions of a cent — same rationale as the SEO ledger). No client-facing money in this module's v1.

### The two RLS decisions this module makes explicitly

- `creative_workflows` is **global, no-RLS, admin-write-only** — it holds pipeline definitions (workflow JSON + model manifests), zero client data; per-tenant duplication would fork the license source of truth. Same class of deliberate exemption as `social_platform_apps` (SMM D-4), narrower than SEO's `search_data_cache` .
- `creative_assets` (shipped with a plain tenant wall in 0031) is **upgraded to the third wall** in this module's migration, in lockstep with moving `CreativeController` into the module. The HTTP contract ( `/api/:t/creative/assets` , used by the studio UI and `creative-grading-trainer` ) is preserved as a stable alias; new surface lives under `/api/:t/modules/creative/*` . This is the one behavior change to shipped code and it ships atomically in CR-01+CR-02 (D-3).



**FIG. 04-1** – Entity map. `creative_render_jobs` is the state-machine spine; `creative_workflows` is the single global source of license truth every job resolves against.

### Key tables

| TABLE                                                         | PURPOSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>creative_assets</code>                                  | <b>Extended in place</b> (additive <code>ALTER TABLE</code> ) — existing 0031/0032 columns untouched; <code>graded_key</code> remains the storage key of the current/primary bytes (exposed as <code>primaryKey</code> ). New columns add <code>kind</code> , <code>source</code> , <code>client_id</code> , <code>project_id</code> , <code>render_job_id</code> , <code>duration_ms</code> , the rights/licensing spine ( <code>license_class</code> , <code>rights jsonb</code> , <code>rights_expires_on</code> , <code>reuse_status</code> , <code>reuse_approval_id</code> , <code>approved_by/at</code> ), the provenance record ( <code>provenance jsonb</code> — workflow/models/backend/prompt hash/seed/derived-from), and search/AI metadata ( <code>checksum_sha256</code> , <code>phash</code> , dual-mode <code>embedding vector(512)</code> , <code>caption</code> , <code>auto_tags</code> , <code>exif</code> , <code>current_version</code> ). Backfill gives existing studio rows <code>source='studio'</code> , <code>license_class='commercial'</code> . |
| <code>creative_asset_versions</code>                          | Superseded versions only — the asset row always holds current bytes so every shipped read path stays correct. New version = copy current into versions, overwrite the asset row, bump <code>current_version</code> , re-run analyze.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>creative_collections / creative_collection_items</code> | <code>kind ('collection','brand_kit')</code> ; brand kit <code>slots jsonb = {logo_asset_ids, palette, fonts, approved_imagery}</code> ; items = ( <code>collection_id</code> , <code>asset_id</code> , <code>position</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>creative_render_jobs</code>                             | The job spine — state machine authoritative here, not in the gateway. See the DDL and lifecycle in §05.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>creative_usage_ledger</code>                            | The metering ledger — one row per dispatched job + true-up (mirrors <code>search_provider_calls</code> ): units (GPU-seconds/images/clip-seconds), <code>cost_usd numeric(12,6)</code> , <code>cost_class</code> (image/video/analyze — separate budget envelopes per locked decision 3). Indexed ( <code>tenant_id</code> , <code>client_id</code> , <code>created_at DESC</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>creative_scopes</code>                                  | The human-set per-client service config (the SEO D-11 pattern, creative-shaped) — see the JSON shape in §05/§06. Effective scope = client row else tenant default else platform default (env). Every dispatch and every scheduled flow consults it; a disabled capability refuses <b>naming the toggle</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| TABLE                | PURPOSE                                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| creative_workflows   | <b>Global, no-RLS, admin-write-only</b> — the single source of truth the router filters on and the license stamp derives from. Workflows are immutable once referenced by a job (new behavior = new version). v1 seeds only commercial-clean workflows active (§06); quarantine rows exist with <code>active=false</code> . |
| automation_approvals | Reused with the origin CHECK widened to include <code>'creative'</code> . Both gates ride it: spend-gate rows ( <code>tool_name='creative.render'</code> ) and reuse-gate rows ( <code>tool_name='creative.approveReuse'</code> ).                                                                                          |

```
CREATE TABLE creative_render_jobs (  
  id uuid PRIMARY KEY,  
  tenant_id uuid NOT NULL REFERENCES companies(id),  
  client_id uuid REFERENCES clients(id),  
  requested_by uuid REFERENCES users(id),          -- human or the OBO automation user  
  requester_module text NOT NULL DEFAULT 'creative'  
  CHECK (requester_module IN ('creative','social','search','webdesk','agent','automation')),  
  capability text NOT NULL CHECK (capability IN  
    ('upscale','generate','edit','bg_remove','relight','analyze',  
     't2v','i2v','flf2v','interpolate','video_upscale')),  
  workflow_key text NOT NULL, workflow_version integer NOT NULL,  
  params jsonb NOT NULL DEFAULT '{}',              -- prompt, size, sliders, masks - client data, RLS-protected  
  input_asset_ids uuid[] NOT NULL DEFAULT '{}',  
  constraints jsonb NOT NULL DEFAULT '{}',          -- {license_floor, max_cost_usd, deny_cloud}  
  status text NOT NULL DEFAULT 'queued' CHECK (status IN  
    ('pending_approval','queued','dispatched','running','uploading',  
     'succeeded','failed','cancelled','refused')),  
  approval_id uuid REFERENCES automation_approvals(id), -- spend gate (one-shot, §07)  
  backend_key text, backend_ref text,               -- stamped at dispatch  
  license_stamp jsonb,                             -- gateway-returned {models,backend}  
  estimated_cost_usd numeric(12,6),  
  actual_cost_usd numeric(12,6),  
  progress integer NOT NULL DEFAULT 0,  
  error text, attempt integer NOT NULL DEFAULT 0,  
  lease_expires_at timestamptz,                    -- reconciler re-drives stale jobs  
  correlation_id text,                             -- MCP call / n8n run id  
  origin_site text NOT NULL, created_at timestamptz NOT NULL DEFAULT now(),  
  updated_at timestamptz NOT NULL DEFAULT now(),  
  queued_at timestamptz, started_at timestamptz, finished_at timestamptz  
);  
CREATE INDEX ix_creative_jobs_active ON creative_render_jobs (tenant_id, status)  
  WHERE status IN ('pending_approval','queued','dispatched','running','uploading');
```

Custom fields & relations

- D17 `customFieldTargets`: `creative_asset`, `creative_collection`.
- **Clients/projects** are the core rows — no duplicates. Delivering an approved asset to a client as a formal deliverable rides the existing `deliverables` flow (seam noted, not new schema).

§05 Creative Render Gateway

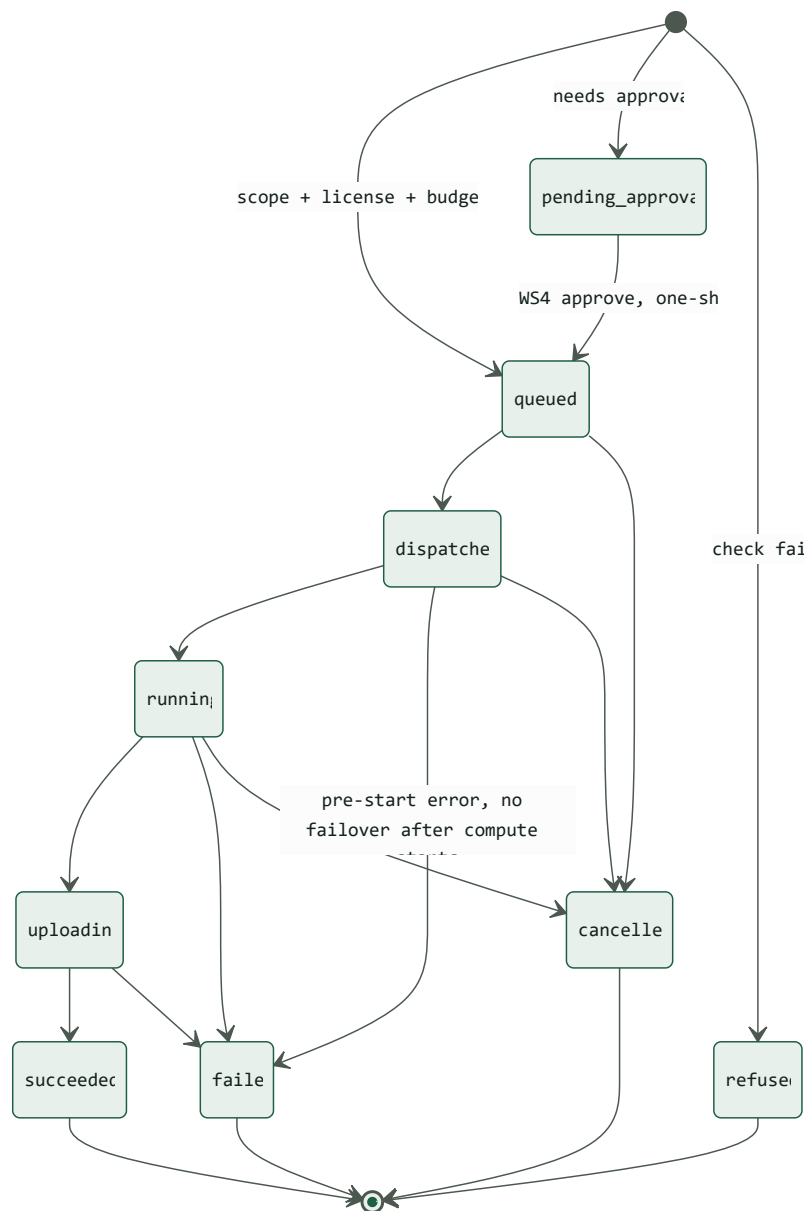
The centerpiece. A separate Go service — mirroring ai-gateway-go's key-custody, egress, and failure-isolation discipline, applied to GPU compute instead of LLM tokens.

Verdict and shape

**A separate Go service**, `render-gateway-go/` (compose service `render-gateway`) — **not** a platform-nest in-process worker, for the same reasons ai-gateway-go is separate: key custody isolation (GPU provider keys never enter the platform process), egress-allowlist enforcement at the dialer, independent failure/restart behavior for long-running jobs, and Go's fitness for concurrent poll/stream workloads. **Platform-nest stays the source of truth for job state**; the gateway is a stateless-ish executor (in-flight leases in memory/Redis; crash ⇒ platform reconciler re-drives by lease timeout, idempotent on `job_id`).

## Job lifecycle

States live on `creative_render_jobs`, not in the gateway:



**FIG. 05-1** – Job state machine. Mid-run failures never failover to another backend (no double spend) – they fail the job for a human decision.

- **Create** (console button or MCP tool): platform-nest validates capability against the effective `creative_scopes`, resolves the workflow ( `key@version`, must be `active` ), computes the **license floor** (client-attributed job  $\Rightarrow$  `commercial|licensed` only), asks the gateway `POST /estimate` for cost, then runs the stop-loss chain (below). Outcome: `queued`, `pending_approval`, or `refused`.
- **Approval (spend gate)**: required when (a) the principal is an automation/agent OBO user — always, D14; (b) `estimated_cost_usd > per_job_approval_threshold` in scope; (c) the resolved route is a **premium backend** (BFL FLUX tier, commercial video API); or (d) capability class `video` for any non-human requester. Approved rows are **one-shot**: dispatch consumes `approval_id` transactionally, replay-proof.
- **Dispatch**: platform-nest  $\rightarrow$  `POST /jobs` on the gateway with the job envelope (capability, workflow + definition hash, params, signed input/output URLs, constraints, callback token). Idempotent on `job_id`. Gateway picks the backend (routing policy below) and leases the job.
- **Execute**: backend-specific — serverless (submit, poll), ComfyUI ( `POST /prompt`, ws/poll `/history`, fetch artifacts), commercial API (submit, poll). Progress callbacks are optional; terminal callback is mandatory and idempotent.
- **Callback**: `POST /api/internal/creative/render-callback` — platform-nest verifies the per-job token, transitions the row (one-way), **creates the output asset or version** with provenance + computed `license_class` (weakest of the license stamp's models U input assets' classes), true-ups the ledger row, emits `creative.render.completed|failed`, and enqueues an `analyze` job (fail-soft).

- **Reconciler:** a periodic module task re-drives `dispatched/running` rows past `lease_expires_at` ( `GET /jobs/:id` first; re-dispatch only for never-started jobs — mid-run failures fail closed to human retry, never silent re-spend).

The RenderBackend abstraction (gateway-side) and routing

**RENDERBACKEND ROUTING**

A cost-ordered, config-driven candidate list per capability (hot-reloadable like the bot's `groups.yaml`). At dispatch the router filters by capability  $\wedge$  license floor (a `commercial` floor excludes every non-commercial route and excludes `licensed` unless the scope has the premium tier)  $\wedge$  `deny_cloud`  $\wedge$  `max_cost_usd`  $\wedge$  circuit-breaker health, then takes the first.

**FAIL-CLOSED STOP-LOSS**

(0) capability enabled in scope  $\rightarrow$  (1) license floor satisfiable  $\rightarrow$  (2) estimate  $\leq$  cap, threshold/premium/automation  $\Rightarrow$  `pending_approval`  $\rightarrow$  (3) month-to-date spend per cost class vs. client  $\rightarrow$  tenant  $\rightarrow$  global cap (image \$200/mo, video \$300/mo, analyze \$10/mo defaults; also fires at 80%)  $\rightarrow$  (4) advisory-lock single-flight per job id  $\rightarrow$  dispatch  $\rightarrow$  true-up on callback.

**LICENSE WALL**

Every workflow's model manifest carries a license class; the router filters routes below the job's floor; the callback stamps the actual models used; an asset's `license_class` = weakest of the derivation chain. `approve_reuse` refuses any chain element  $\notin$  {commercial, licensed}. There is no admin override.

```
// Design sketch – capability-based so drivers can be partial (SearchDataProvider's sibling).
type RenderBackend interface {
    Key() string // 'replicate' | 'fal' | 'runpod-serverless' | 'comfy-selfhost'
               // | 'bfl-api' | 'video-api' | 'analyze-cpu' | 'echo' (tests)

    Capabilities() []Capability
    // License class of what THIS backend runs for a given workflow/model – the router's filter input.
    LicenseClass(workflow WorkflowRef) LicenseClass // commercial | licensed | noncommercial
    EstimateCostUsd(job JobSpec) (float64, error) // consulted BEFORE dispatch (stop-loss + approval threshold)
    Submit(ctx context.Context, job JobSpec) (BackendRef, error)
    Poll(ctx context.Context, ref BackendRef) (JobUpdate, error)
    Cancel(ctx context.Context, ref BackendRef) error
}
```

**Failover:** pre-start errors (429/5xx before compute starts) advance to the next candidate; **mid-run failures never failover** (no double spend) — they fail the job with the backend's error for a human decision. Per-tenant concurrent-job caps and a global concurrency cap are gateway config.

ComfyUI workflows as versioned JSON

The unit of pipeline definition is a `creative_workflows` row: the exact ComfyUI graph JSON (or API template), a params schema (prompt, image refs, denoise, creativity/HDR sliders mapped to denoise-strength + ControlNet weight), and the **model manifest with license classes**. Platform-nest resolves `key@version` and sends the definition hash; the gateway verifies hash before execution (a tampered workflow refuses).

**Seeded v1 set:** `upscale-clarity@1`, `upscale-precision@1`, `relight-iclight1@1`, `gen-qwen@1`, `gen-sd-xl@1`, `gen-zimage@1`, `edit-qwen@1`, `bgrremove-birefnet@1`, `analyze-clip-blip@1`, `t2v-wan22@1`, `i2v-wan22@1`, `flf2v-wan22@1`, `interp-rife@1`, `vupscale-esrgan@1`; premium: `gen-flux-bfl@1` (api engine). **Quarantined-inactive** (documentation of the boundary, not tribal knowledge): `upscale-supir@0`, `relight-iclight2@0`.

The ledger powers the Jobs & Usage panel, tenant rollups, and the `creative.render_cost.month` metric that feeds billing. Manual override = `creative:provider:admin` + audit.

§06 Backend / Model / License Matrix

The locked hybrid, made operational. Default pipeline = commercial-clean only; the premium tier is a scope flag + permission, never a default.

| CAPABILITY          | MODEL / WORKFLOW                          | LICENSE CLASS                              | BACKEND(S), COST-ORDERED                              | NOTES                                     |
|---------------------|-------------------------------------------|--------------------------------------------|-------------------------------------------------------|-------------------------------------------|
| Upscale (creative)  | clarity-upscaler (SDXL + ControlNet-Tile) | <code>commercial</code> (AGPL tool — OQ-8) | replicate (~\$0.019/img) $\rightarrow$ comfy-selfhost | THE Magnific replacement; P0 spike target |
| Upscale (precision) | Real-ESRGAN · DiffBIR                     | <code>commercial</code> (BSD/Apache)       | replicate/fal $\rightarrow$ comfy-selfhost            | fast/faithful mode                        |

| CAPABILITY                          | MODEL / WORKFLOW                                                                                               | LICENSE CLASS                       | BACKEND(S), COST-ORDERED                               | NOTES                                                                                                             |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Relight                             | IC-Light <b>V1</b>                                                                                             | <b>commercial</b>                   | comfy-selfhost → fal                                   | <b>V2 is NC — quarantined</b>                                                                                     |
| Generate                            | Qwen-Image · SDXL · Z-Image-Turbo · FLUX schnell/klein                                                         | <b>commercial</b> (Apache/OpenRAIL) | fal/replicate serverless → comfy-selfhost              | Qwen = text-rendering strength; Z-Image = cheap/fast                                                              |
| Generate (hero)                     | <b>FLUX dev-class</b> via BFL/fal hosted API                                                                   | <b>licensed</b>                     | bfl-api / fal                                          | <b>Premium tier only:</b> scope <code>premium.flux</code> + <code>creative:render:premium</code> + spend approval |
| Edit (instruction)                  | Qwen-Image-Edit                                                                                                | <b>commercial</b> (Apache)          | fal/replicate → comfy-selfhost                         | default editing engine                                                                                            |
| BG removal                          | <b>BiRefNet</b> (MIT)                                                                                          | <b>commercial</b>                   | analyze-cpu / replicate                                | <b>RMBG is NC — never</b>                                                                                         |
| Inpaint/outpaint/style/structure    | ControlNet + IP-Adapter (SDXL)                                                                                 | <b>commercial</b>                   | comfy-selfhost → serverless                            | Magnific-style sliders live here                                                                                  |
| T2V / I2V / FLF2V                   | <b>Wan 2.2</b> (5B/14B)                                                                                        | <b>commercial</b> (Apache)          | runpod-serverless ComfyUI (~\$0.03–0.10/clip compute)  | anchor; FLF2V = the A→B morph requirement                                                                         |
| Video (hero / synced audio)         | <b>Veo</b> · Kling/Runway                                                                                      | <b>licensed</b> (commercial API)    | video-api via fal/replicate                            | <b>Premium tier</b> , \$100–300/mo envelope, per-clip approval                                                    |
| Interpolate / video upscale         | RIFE · Real-ESRGAN                                                                                             | <b>commercial</b>                   | comfy-selfhost / serverless                            | post-pipeline for masters                                                                                         |
| Analyze (embed/caption/tag)         | CLIP ViT-B/32 + BLIP-base                                                                                      | <b>commercial</b>                   | <b>analyze-cpu</b> (local container, ~\$0) → replicate | fail-soft ingest enrichment                                                                                       |
| <b>Quarantined — ships disabled</b> | SUPIR · IC-Light V2 · RMBG · SVD · FLUX dev free weights · HunyuanVideo (geo-caveat) · LTX (license ambiguous) | <b>noncommercial / caveated</b>     | none active                                            | Rows exist <code>active=false</code> so the boundary is visible                                                   |

FLUX LICENSING – THE BIGGEST CLIENT-FACING TRAP

All FLUX *dev-class* weights are non-commercial. For client work either (a) buy BFL's self-serve commercial license (their "Professional/agency" tier covers up to ~3 clients, then per-client fees — quote-gated, OQ-1), or (b) call the hosted BFL/fal/Replicate API where the license is bundled in the price. **Never ship FLUX dev output to a paying client on the free weights.**

AGPL / GPL POSTURE (OQ-8, COUNSEL MEMO PENDING)

clarity-upscaler is AGPL-3.0 — internal-use tool, outputs delivered, posture documented for sign-off. ComfyUI (GPL-3.0) runs headless behind the gateway at arm's length over HTTP only — GPL never links into our proprietary code. LTX-Video's license is genuinely ambiguous across sources; confirm with Lightricks before any commercial use. HunyuanVideo excludes EU/UK/S.Korea and carries a 100M-MAU cap — stays out of v1 regardless.

§07 AI Design

Local-first task routing, the CLIP/BLIP analyze-job pattern that avoids the cross-modal embedding-space pitfall, and the two-gate WS4 spine.

| TASK                                                   | WHERE IT RUNS                                                                                      | TRIGGER                                | NOTES                                                                                                                                                                                                                                      |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prompt expansion / negative-prompt / style suggestions | <b>ai-gateway-go</b> <code>/complete</code> — Hermes default, Claude flag for client-facing copy   | Generate/Edit canvas                   | Never a direct vendor call; gateway DLP + caps apply                                                                                                                                                                                       |
| Alt-text + tag normalization (post-BLIP cleanup)       | ai-gateway <code>/complete</code> (Hermes)                                                         | analyze pipeline                       | Draft metadata only                                                                                                                                                                                                                        |
| Image captioning                                       | <b>BLIP-base as an analyze render job</b> on the CPU worker backend                                | asset ingest (event-driven, fail-soft) | NOT via gateway <code>/media</code> — local-first, \$0, batch-friendly                                                                                                                                                                     |
| Visual/semantic embeddings                             | <b>CLIP ViT-B/32</b> as the same <code>analyze</code> job → <code>creative_assets.embedding</code> | ingest + on new version                | <b>Pitfall made explicit:</b> text→image search must encode the query with the CLIP text encoder (same job capability), NOT ai-gateway <code>/embed</code> (nomic-space ≠ CLIP-space). <code>/embed</code> is never used for asset search. |
| Similar-asset search + dedup                           | Postgres: checksum exact → pHash near-dup → pgvector cosine                                        | Library UX, pre-save dedup hint        | Dual-mode column (pgvector else <code>float8[]</code> + app-side cosine — the WS8 store pattern; OQ-7 shared with SEO)                                                                                                                     |
| Keyword search                                         | Postgres FTS over <code>name/caption/auto_tags</code>                                              | Library                                | No extra engine                                                                                                                                                                                                                            |

D9 HYGIENE (SAME REASONING AS SEO D-7)

`creative_assets.embedding` is an *operational feature column* (search/dedup over module rows), not a knowledge store. Anything retrieval-shaped (brand-voice corpora, style guides) stays in WS8 via `knowledge.search`. The existing `creative-grading-trainer` (learned house-look, Image Studio phase-2) is unchanged by this design — the 3D-LUT seam stays the attach point.

The two-gate WS4 spine

- 1. **Spend gate (before GPU money):** \$05 choke point. Automation/agent principals **always** suspend into `automation_approvals` (D14 — MCP render tools are `write:true`); humans suspend above the per-job threshold or on premium routes. Execution consumes the one-shot approvalId with payload-hash match.
- 2. **Reuse gate (before an asset can leave the department):** a generated/edited asset is born `reuse_status='private'`. "Submit for reuse" → `pending_approval` + an approvals-inbox item showing preview + **the full license chain**; approval requires every element of the provenance chain ∈ {commercial, licensed} — the service refuses otherwise (structural rule; there is no override flag). Approved assets become visible to cross-dept consumers (\$08) and deliverable to clients; `revoked` removes them from the shared surface and emits an event.

Creative capabilities as MCP tools (fills the reserved `image.enhance` seam)

Registered via `ModuleContract.mcpTools` (hub-aggregated). Reads `minAssurance: 'low'`; every GPU-spending tool is `write:true` (spending money is a mutation — SEO D-5), so automation principals hit the D14 gate by construction.

| TOOL                                                                                                                                          | KIND                   | IMPACT                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------------------|
| <code>creative.searchAssets</code> / <code>getAsset</code> / <code>listCollections</code> / <code>listJobs</code> / <code>usageSummary</code> | read                   | —                                     |
| <code>creative.enhanceImage</code>                                                                                                            | GPU spend              | medium (budget-checked)               |
| <code>creative.generateImage</code> / <code>editImage</code> / <code>removeBackground</code>                                                  | GPU spend              | medium                                |
| <code>creative.generateVideo</code> / <code>animateImage</code> (t2v/i2v/f1f2v)                                                               | GPU spend, video class | high → always suspends for automation |

| TOOL                                 | KIND                       | IMPACT                                                  |
|--------------------------------------|----------------------------|---------------------------------------------------------|
| <code>creative.submitForReuse</code> | starts the reuse approval  | <span>Low</span> (no side effect beyond the inbox item) |
| <code>creative.captionAssets</code>  | cheap job (re-run analyze) | <span>Low</span>                                        |

## §08 Console UX

Dept Creatives on the two-level dept-interface-template — universal spine plus two craft groups, growing from one (production vs. supply is the natural split).

| GROUP                         | SUB-TABS (ROUTE UNDER <span>/DEPARTMENTS/[DEPTID]/</span> )                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Studio</b><br>(production) | Image Studio ( <span>studio</span> — PROTOTYPED, unchanged) · Generate & Edit ( <span>generate</span> ) · Upscale ( <span>upscale</span> ) · Video ( <span>video</span> ) |
| <b>Library</b> (supply)       | Assets ( <span>assets</span> — exists, upgraded to the DAM surface) · Brand Kits ( <span>brand-kits</span> ) · Jobs & Usage ( <span>jobs</span> )                         |

Home = command-center KPIs from rollups (jobs this month, MTD render spend vs. caps, assets approved for reuse, rights expiring ≤30d) + activity feed + launchers (Shared Drive, Figma, Claude). The My-work rail is inherited unchanged.

### Button capability matrix

Every ●/● action is additionally gated by the client's creative\_scopes config: a disabled capability renders the button disabled and names the missing toggle.

● local/\$0 – client-side studio, own-data search, drafts
● GPU-metered – serverless spend; scope + ledger stop-loss

● PREMIUM – FLUX licensed tier / commercial video API, explicit paid opt-in

● WS4 approval – human decision before execution

| CONSOLE ACTION                                    | TAB             | NEEDS                        | GATE                                                                                    |
|---------------------------------------------------|-----------------|------------------------------|-----------------------------------------------------------------------------------------|
| Auto-correct / grade / batch-export / Save to ERP | Image Studio    | <span>●</span>               | permission only (shipped behavior)                                                      |
| Creative upscale (clarity recipe, sliders)        | Upscale         | <span>●</span>               | budget stop-loss                                                                        |
| Precision upscale / relight (IC-Light V1)         | Upscale         | <span>●</span>               | budget stop-loss                                                                        |
| Batch upscale (n8n-driven)                        | Upscale         | <span>●</span>               | <span>●</span> above threshold; always <span>●</span> for automation                    |
| Generate image (Qwen / SDXL / Z-Image)            | Generate & Edit | <span>●</span>               | budget stop-loss                                                                        |
| Prompt expansion / style assist                   | Generate & Edit | <span>●</span> (Hermes)      | draft only                                                                              |
| Instruction edit / inpaint / outpaint / bg-remove | Generate & Edit | <span>●</span>               | budget stop-loss                                                                        |
| <b>FLUX hero generation</b>                       | Generate & Edit | <span>●</span>               | scope <span>premium.flux</span> + <span>creative:render:premium</span> + <span>●</span> |
| Wan T2V / I2V / first+last-frame morph            | Video           | <span>●</span>               | video budget; <span>●</span> above threshold                                            |
| <b>Veo / Kling hero clip (syncd audio)</b>        | Video           | <span>●</span>               | scope <span>premium.video_api</span> + <span>●</span> per clip                          |
| Interpolate (RIFE) / video upscale                | Video           | <span>●</span>               | video budget                                                                            |
| Search / find-similar / dedup check               | Assets          | <span>●</span> (post-ingest) | —                                                                                       |

| CONSOLE ACTION                                     | TAB          | NEEDS                                                 | GATE                                                                        |
|----------------------------------------------------|--------------|-------------------------------------------------------|-----------------------------------------------------------------------------|
| Re-run caption/tags (analyze)                      | Assets       | <span>●</span> / <span>●</span> (CPU worker<br>≈ \$0) | —                                                                           |
| Edit rights / license metadata                     | Assets       | <span>●</span>                                        | <code>creative:asset:write</code>                                           |
| <b>Submit / approve asset for cross-dept reuse</b> | Assets       | <span>●</span>                                        | <span>●</span> (reuse gate; license chain must be clean)                    |
| Revoke reuse (rights expiry etc.)                  | Assets       | <span>●</span>                                        | <code>creative:asset:approve_reuse</code>                                   |
| Assemble / edit brand kit                          | Brand Kits   | <span>●</span>                                        | <code>creative:asset:write</code>                                           |
| Copy rendition URL (per-network crops)             | Assets       | <span>●</span>                                        | signed URL, read permission                                                 |
| View job queue / cancel job / view ledger          | Jobs & Usage | <span>●</span>                                        | <code>creative:ledger:read</code>                                           |
| Raise caps / edit client scope config              | Jobs & Usage | <span>●</span>                                        | <code>creative:scope:write</code> /<br><code>creative:provider:admin</code> |

Everything ● plus the ● **upscale line ships value in P1** — the Magnific replacement does not wait for the generative canvas.

## §09 ERP Integration Points

Reuse over invention — one new deployable (the Render Gateway); every other plane is an existing platform mechanism applied to a new resource set.

| SUBSYSTEM                   | INTEGRATION (CONCRETE)                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| platform-nest               | <code>ModuleContract</code> key <code>creative</code> ; controller <code>@Controller("api/:tenantId/modules/creative")</code> (hr/search convention) + the preserved legacy alias <code>api/:tenantId/creative/assets</code> (studio UI + trainer compat); <code>ModuleEnabledGuard</code> ; enablement via <code>enabled_modules</code> OR active <code>service_assignment</code> |
| BFF contract                | New section in <code>FRONTEND-BFF-CONTRACT.md</code> : <code>/api/:t/modules/creative/{assets,assets/:id/versions,collections,brand-kits,scopes,jobs,jobs/:id,ledger,search,renditions/sign}</code> — shapes canonical in <code>platform-ui/src/lib/creative.ts</code> (extends the existing file)                                                                                 |
| Creative Render Gateway     | The §05 service — the one new deployable. Compose service <code>render-gateway</code> ; GPU keys env→OpenBao; egress allowlist; OTel                                                                                                                                                                                                                                               |
| ai-gateway-go               | <code>/complete</code> for prompt/copy assist (Hermes default, Claude flag). NOT used for CLIP/BLIP (§07)                                                                                                                                                                                                                                                                          |
| mcp-hub                     | <code>creative.*</code> tools via <code>mcpTools</code> aggregation (§07); closes the hub's reserved Magnific <code>image.enhance</code> deferral                                                                                                                                                                                                                                  |
| WS8 knowledge / pgvector    | Embedding column is operational (dual-mode); brand/style corpora stay WS8-owned; OQ-7 = pgvector in platform DB images (shared with SEO OQ-8)                                                                                                                                                                                                                                      |
| WS4 approvals               | <code>automation_approvals.origin</code> widened with <code>'creative'</code> ; spend gate + reuse gate both land in the existing inbox; one-shot approvalId consumption at dispatch                                                                                                                                                                                               |
| automation (n8n)            | Flows in §10; backbone rule — n8n orchestrates, MCP accesses, zero logic in workflows                                                                                                                                                                                                                                                                                              |
| Event backbone              | <code>creative.asset.ingested</code> , <code>creative.render.completed failed awaiting_approval</code> , <code>creative.budget.threshold</code> , <code>creative.asset.reuse_approved reuse_revoked rights_expiring</code> → bell + n8n bridge                                                                                                                                     |
| Files / Shared Drive (WS11) | Asset bytes in the existing storage backend; approved client deliverables mirrored to the client's Drive folder; video default = local storage + Drive mirror                                                                                                                                                                                                                      |
| imgproxy                    | New Zone A sidecar (MIT, Go/libvips): signed, expiring rendition URLs — no rendition rows (D-8)                                                                                                                                                                                                                                                                                    |
|                             |                                                                                                                                                                                                                                                                                                                                                                                    |

| SUBSYSTEM                       | INTEGRATION (CONCRETE)                                                                                                                                                                                                                                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SMM / SEO / WebDesk (consumers) | Read surface = <b>approved assets only</b> : GET /api:t/modules/creative/assets?reuseStatus=approved + POST ... /renditions/sign . Consumers never touch creative_* tables directly (the third wall enforces this structurally). GPU spend requested by siblings books once with requester_module attribution (D-7) |
| Rollups (D12)                   | creative.renders.month · creative.render_cost.month (money, isMonetary — feeds billing) · creative.assets.added.month · creative.assets.approved_total · creative.rights.expiring_30d                                                                                                                               |
| Cerbos                          | New resource policies (§11); asset byte access keeps riding the existing file policy; lib/rbac.ts capability mirror                                                                                                                                                                                                 |
| observability (WS9)             | OTel spans: job dispatch/poll/callback with attrs (backend, model, capability, cost_usd); per-backend circuit-state + GPU-seconds gauges; ledger-vs-cap gauge; fail-soft OTEL_ENABLED                                                                                                                               |

§10 Automation Flows

Thin orchestrations calling creative.\* MCP tools (impact-gated automatically). JSON lives in automation/workflows/ , kebab-named.

| FLOW                 | SCHEDULE / TRIGGER                             | MCP CALLS                                                                                                    | PHASE       |
|----------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------|
| cr-ingest-analyze    | event<br>creative.asset.ingested               | creative.captionAssets (analyze job: CLIP+BLIP+pHash)                                                        | P3          |
| cr-batch-upscale     | manual trigger / dropbox folder convention     | creative.enhanceImage per item (each budget-checked; batch > threshold suspends)                             | P1          |
| cr-render-notify     | events<br>creative.render.complete<br>d failed | notify requester + on approved-deliverable jobs mirror to Drive                                              | P1          |
| cr-ledger-guard      | daily                                          | creative.usageSummary → threshold events → notify / pause queues                                             | P1          |
| cr-rights-expiry     | weekly                                         | list assets rights_expires_on ≤ 30d → creative.asset.rights_expiring events → notify + auto-revoke at expiry | P3          |
| cr-agent-asset-brief | WS8 agent goal (post-P2)                       | creative.generateImage drafts → all land as private assets + approvals inbox                                 | P4-adjacent |

§11 Trust & Security

Nothing new to invent — every control reuses an existing platform mechanism, applied to GPU keys, generative pixels, and their license provenance.

**KEY CUSTODY**

GPU/model provider keys (RunPod, fal, Replicate, BFL, Veo, Kling): render gateway env → OpenBao target-state; never in platform-nest, platform-ui, or n8n. Storage access: per-job signed URLs only. imgproxy signing key: imgproxy + platform-nest env.

**RLS**

All creative\_\* tenant tables: FORCE-RLS, third-wall predicate, fail-closed empty-set semantics. creative\_assets policy upgraded in the same migration (D-3). The single exemption is creative\_workflows (global, admin-write-only — D-4).

**CERBOS RESOURCES**

resource\_creative\_asset (read/write/set\_rights/submit\_reu  
resource\_creative\_render\_job (read/submit/submit\_video/subm  
resource\_creative\_scope , resource\_creative\_ledger , resource\_creative\_workfl  
ow (platform-admin only).

**LICENSE WALL (STRUCTURAL)**

Router filters routes below the job's license floor; callback stamps the actual models used; asset license\_class = weakest of the derivation chain; approve\_reuse and deliverable paths refuse any chain element ∉ {commercial, licensed}. No admin override exists.

MONEY SAFETY

Single dispatch choke point (scope → license → estimate/approval → MTD caps per cost class → advisory-lock → dispatch → true-up); one-shot approvald with payload-hash match; no cross-backend failover after compute starts; daily `cr-  
ledger-guard` as the independent watchdog.

EGRESS CONTROL + PRIVACY

DialContext-enforced allowlist of backend hosts; per-job egress audit (tenant/client hashes, backend, bytes); opaque job ids in anything a provider sees; `allow_cloud_gpu:false` scope pins a client to self-host routes (OQ-5).

UNTRUSTED COMPUTE

Outputs from any backend are size-capped, content-type-validated, and re-encoded on ingest; workflow definition hash verified before execution; callback token is per-job and single-purpose.

AUDIT

Ledger row per dispatch, Cerbos decision audit, hub JSONL for every MCP call, approvals decisions, egress audit lines — nothing new to invent.

Module permissions declared in the contract: `creative:asset:read|write`, `creative:asset:approve_reuse`, `creative:render:submit`, `creative:render:video`, `creative:render:premium`, `creative:scope:write`, `creative:ledger:read`, `creative:provider:admin`, `creative:workflow:admin`.

§12 Rollout & Ticket Decomposition

Cost-honest phases — Phase 0 kills the Magnific bleed with zero platform code; P0 contracts → P1 upscale → P2 image generate/edit → P3 DAM + cross-dept supply → P4 video. 27 tickets (CR-00–CR-26), /army-ready.

Tiers per the agent-army standard; **model = seat default unless flagged**. ⚡ = touches a contract (schema/API/policy/license boundary) → QA gate + architect design-review on the diff. Registration in `MODULES.md` as `creative · 0.0.0 · PLANNED` happens on approval of this doc; the first merged ticket flips it to `IN PROGRESS` + CHANGELOG entry.

P0

Stop the Magnific bleed (this week — below mobilization threshold, run inline)

CR-00: clarity-upscaler on Replicate spike — team-usable invocation path, preset notes, cost log, a 10-image parity set vs. Magnific output.

GATE → 10 reference images upscaled at ~\$0.02/run; parity notes written; team told to stop buying Magnific credits.

P0

Contracts

CR-01–05: module migration + third-wall retrofit, ModuleContract + controller move, Cerbos policies, Render Gateway skeleton, config plumbing.

GATE → module registers, RLS proven, gateway echo-loop round-trips end to end.

P1

Upscale — the Magnific replacement, through the platform

CR-06–11: router + metering choke point, serverless backend driver, job UX-backend, Upscale + Jobs&Usage tabs, notification wiring, parity bake-off.

GATE → per-image platform cost ≤ \$0.03; parity notes signed off; Magnific subscription cancellable.

P2

Image generation + editing

CR-12–16: ComfyUI serverless backend (QA-gated), the two gates (spend + reuse), Generate & Edit canvas, prompt assist, P2 e2e.

GATE → generate → edit → submit-for-reuse → approve (license chain shown) → visible in the shared surface.

### P3

#### DAM search/tagging + cross-dept supply

CR-17–22: analyze pipeline, DAM search surface, Library UX v2 + Brand Kits, imgproxy sidecar (QA-gated), cross-dept consumption seams, rights-expiry flow + docs.

GATE → a served sibling company's SMM console lists only approved creative assets of its own tenant; expiry fixture auto-revokes correctly.

### P4

#### Video

CR-23–26: Wan 2.2 video workflows, commercial video-API driver, Video tab UX, full-stack e2e verification.

GATE → scripted e2e green on the live dev stack (upscale → generate → reuse-approve → SMM pulls rendition → video clip → ledger reconciliation); DEV-VERIFIED criteria documented.

| #       | TICKET                                                                                                                                                                                                                                  | TIER              | MODEL             | PHASE   | DEPS     |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|---------|----------|
| CR-00   | clarity-upscaler on Replicate spike: invocation path + preset notes + cost log + 10-image parity set vs. Magnific                                                                                                                       | medior            | default           | Phase 0 | —        |
| CR-01 ⚡ | Migration <code>0036_module_creative.sql</code> : extend <code>creative_assets</code> + third-wall policy replacement, all \$04 tables + seeds, <code>automation_approvals</code> origin widen, dual-mode embedding column              | senior-db         | opus·medium       | P0      | —        |
| CR-02 ⚡ | <code>creative</code> ModuleContract + NestJS module: registry, permissions, guard, uiManifest, <code>CreativeController</code> moved in (legacy path aliased), DAM CRUD, rollup providers                                              | senior-be         | default           | P0      | CR-01    |
| CR-03 ⚡ | Cerbos policies × 5 + derived-roles wiring + policy tests + <code>lib/rbac.ts</code> mirror                                                                                                                                             | medior            | default           | P0      | CR-02    |
| CR-04 ⚡ | <b>Render Gateway skeleton</b> ( <code>render-gateway-go/</code> ): job-envelope contract, jobs/estimate/cancel/health endpoints, callback tokens, lease store, echo backend, egress-allowlist dialer, OTel, compose entry              | senior-be         | default           | P0      | CR-01    |
| CR-05   | Config plumbing: gateway env (provider keys, allowlist, concurrency caps), platform env (caps per cost class, thresholds), <code>.env.example.s</code>                                                                                  | junior            | default           | P0      | CR-04    |
| CR-06 ⚡ | <b>Router + metering choke point</b> : RenderBackend registry, cost-ordered routing, circuit breakers, pre-start-only failover, platform-side stop-loss chain + ledger writes + threshold events                                        | senior-be         | opus·medium       | P1      | CR-04    |
| CR-07   | Serverless backend driver (Replicate + fal): clarity-upscaler, Real-ESRGAN, BiRefNet, IC-Light V1 endpoints; per-call cost accounting; CR-00 presets codified                                                                           | senior-be         | default           | P1      | CR-06    |
| CR-08   | Upscale/relight job UX-backend: submit/list/cancel endpoints, version-vs-new-asset semantics, input signed-URL issuance                                                                                                                 | medior            | default           | P1      | CR-06,02 |
| CR-09 ⚡ | Console P1: Upscale tab (submit + sliders + before/after compare) + Jobs & Usage tab + <code>lib/creative.ts</code> BFF types + toolkit groups update                                                                                   | senior-fe         | default           | P1      | CR-08,03 |
| CR-10   | Events → notifications wiring (all \$09 event types) + flows <code>cr-render-notify</code> , <code>cr-ledger-guard</code> , <code>cr-batch-upscale</code>                                                                               | junior            | default           | P1      | CR-08    |
| CR-11   | P1 verification: Magnific-parity bake-off, cost-per-image report, MODULES.md → <code>IN PROGRESS</code> + CHANGELOG                                                                                                                     | medior            | default           | P1      | CR-09,10 |
| CR-12 ⚡ | <b>ComfyUI serverless backend</b> (RunPod serverless worker template, OQ-4): workflow-JSON execution, model-cache volume strategy, gen/edit/bgremove workflows seeded; <b>QA gate mandatory</b> (egress/creds/untrusted-compute ingest) | senior-integrator | default · QA gate | P2      | CR-06    |

| #          | TICKET                                                                                                                                                                                                         | TIER              | MODEL             | PHASE | DEPS     |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-------|----------|
| CR-13<br>⚡ | <b>The two gates:</b> spend-gate WS4 integration + reuse gate (submit/approve/revoke, license-chain verification) + FLUX premium tier (BFL/fal api-engine workflow behind scope + permission)                  | senior-be         | opus · high       | P2    | CR-06,03 |
| CR-14      | <b>Generate &amp; Edit canvas</b> (v1 form-canvas: prompt + presets + reference image + mask-lite; model picker limited to allowed routes; premium visibly badged)                                             | senior-fe         | default           | P2    | CR-09,13 |
| CR-15      | Prompt assist + style/preset library (ai-gateway Hermes; per-brand-kit style snippets)                                                                                                                         | medior            | default           | P2    | CR-14    |
| CR-16      | P2 e2e on the dev stack: generate → edit → submit-for-reuse → approve (license chain shown) → visible in the shared surface                                                                                    | medior            | default           | P2    | CR-13,14 |
| CR-17      | <b>Analyze pipeline:</b> CPU worker backend container (CLIP + BLIP + pHash), <code>analyze-clip-blip@1</code> workflow, ingest-event trigger, dual-mode embedding writes, fail-soft                            | medior            | default           | P3    | CR-06    |
| CR-18      | <b>DAM search surface:</b> keyword FTS + similar_to + similar_text (CLIP text encoder) + dedup endpoint (checksum → pHash → cosine)                                                                            | medior            | default           | P3    | CR-17    |
| CR-19      | Library UX v2 + Brand Kits tab: filters, rights badges + expiry warnings, similar-to affordance, brand-kit slots editor, thumbnail rendition wiring                                                            | senior-fe         | default           | P3    | CR-18,20 |
| CR-20<br>⚡ | <b>imgproxy sidecar:</b> compose service, signed expiring URLs, <code>POST /renditions/sign</code> BFF endpoint, per-network crop presets; <b>QA gate mandatory</b> (URL forgery, key handling, no open proxy) | senior-integrator | default · QA gate | P3    | CR-02    |
| CR-21      | Cross-dept consumption seams: approved-only listing endpoint hardening, SMM attach-flow contract note (supersedes SMM-20's ledger booking per D-7), BFF rows, <code>deptToolkits</code> cross-links            | medior            | default           | P3    | CR-13,20 |
| CR-22      | <code>cr-rights-expiry</code> flow + revoke path + docs/registration: MODULES.md bump, CHANGELOG, BFF-contract rows, runbook stub                                                                              | junior            | default           | P3    | CR-19    |
| CR-23      | <b>Wan 2.2 video workflows</b> on serverless GPU: t2v/i2v/flf2v + RIFE/Real-ESRGAN post-pipeline, duration/resolution preflight estimator, video storage + Drive mirror                                        | senior-integrator | default           | P4    | CR-12    |
| CR-24      | <b>Commercial video API driver</b> (video-api via fal/Replicate: Veo + Kling/Runway, vendor bake-off in-ticket): per-clip approval above threshold, monthly envelope enforcement                               | senior-be         | default           | P4    | CR-13,23 |
| CR-25      | <b>Video tab UX:</b> t2v/i2v/flf2v submit forms, clip review player, version/master handling, premium badging, per-clip approval states                                                                        | senior-fe         | default           | P4    | CR-23,24 |
| CR-26      | P4 e2e + full-stack verification (upscale → generate → reuse-approve → SMM pulls rendition → video clip → ledger reconciliation) + Playwright suite + DEV-VERIFIED criteria                                    | medior            | default           | P4    | all      |

#### COUNT BY TIER (27 TICKETS CR-00-CR-26)

senior-db 1 (CR-01) · senior-be 6 (CR-02, 04, 06, 07, 13, 24) · senior-fe 4 (CR-09, 14, 19, 25) · senior-integrator 3 (CR-12, 20, 23) · medior 10 (CR-00, 03, 08, 11, 15, 16, 17, 18, 21, 26) · junior 3 (CR-05, 10, 22). **Opus flags: 3** — CR-01 (medium, RLS retrofit on a live shipped table), CR-06 (medium, money-safety concurrency across two services), CR-13 (high, the authz-critical approve-execute-replay surface plus the license wall). **QA gates:** CR-01 (RLS), CR-06 (money), CR-12 (egress/untrusted compute), CR-13 (authz/license), CR-20 (signed URLs). Concurrency: respect the 1–2 agent cap; safe early pairs are (CR-03 || CR-04) and (CR-07 || CR-08); CR-06 and CR-13 must each run alone (they define the spines everything after consumes). CR-00 is below the mobilization threshold — run it inline, not via /army.

§13 Open Questions

Owner decisions — none block P0/P1; several are decided in-ticket at the phase that needs them.

| #    | QUESTION                                                                                                                                                                                           | BLOCKS                                                       | DEFAULT IF UNANSWERED                                                                                                       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| OQ-1 | <b>FLUX license procurement:</b> buy BFL's self-serve commercial license (agency tier, ~3 clients then per-client fees, quote-gated) vs. hosted-API-only (license bundled per call)?               | Nothing structural (premium tier ships API-first either way) | <b>Hosted API only;</b> revisit when FLUX volume makes per-call pricing worse than the license                              |
| OQ-2 | <b>Owned-GPU tripwire:</b> at what sustained spend does a 24 GB box get bought? Proposed: serverless+rented GPU spend > \$250/mo for 3 consecutive months, OR a data-residency client lands (OQ-5) | Nothing in v1 (posture locked serverless-first)              | Adopt the proposed tripwire; ledger rollups make it measurable                                                              |
| OQ-3 | <b>Video-API vendor pick</b> + envelope split inside the locked \$100–300/mo (Veo for audio vs. Kling/Runway for hero)                                                                             | CR-24                                                        | Bake-off at CR-24; Veo weighted for synced audio per the lock                                                               |
| OQ-4 | <b>ComfyUI serverless hosting template:</b> custom worker image vs. community RunPod template; model-cache volume strategy                                                                         | CR-12                                                        | Custom slim image + network-volume model cache (reproducibility over convenience)                                           |
| OQ-5 | <b>Confidential-client tier:</b> is <code>allow_cloud_gpu:false</code> needed for a real client in v1, pulling the rented dedicated ComfyUI box forward?                                           | Backend priority order only                                  | Ship the scope flag; activate the dedicated box only when such a client exists                                              |
| OQ-6 | <b>Canvas depth:</b> embed InvokeAI's Apache-2.0 layered canvas vs. keep growing our own form-canvas (miniPaint/Filerobot MIT base)?                                                               | Post-P2 editor depth only                                    | Own-canvas path (matches the "own the UI, engine behind BFF" doctrine); prototype InvokeAI embed only if editors hit a wall |
| OQ-7 | <b>pgvector extension in the platform DB images</b> (dual-mode fallback keeps plain-PG tests green) — shared decision with SEO OQ-8                                                                | CR-01/17 vector mode                                         | Add to dev/VPS images                                                                                                       |
| OQ-8 | <b>Counsel memo:</b> AGPL posture for clarity-upscaler (internal-use tool, outputs delivered), GPL arm's-length for ComfyUI, and the \$06 quarantine list sign-off                                 | Client-facing output at scale (build proceeds)               | Proceed on the documented arm's-length/internal-use posture; get sign-off before P2 client volume                           |

§14 Decision Log

Four decisions were locked upstream by the owner; this design adds thirteen more, none of which overturn the lock.

LOCKED UPSTREAM (FOUNDATION DECISIONS BLOCK, 2026-07-23 - NOT RELITIGATED HERE)

Serverless/rent-by-second GPU first, owned box later · hybrid image licensing (commercial-clean default, FLUX behind an explicit paid opt-in tier) · hybrid video (Wan 2.2 OSS on rented GPU + \$100–300/mo commercial-API budget for Veo/Kling/Runway) · build-light DAM on our own stack.

| #   | DECISION                                                                                                                                                                                  | WHY                                                                                                                                        |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| D-1 | Render Gateway = <b>separate Go service</b> ( <code>render-gateway-go</code> ); platform-nest owns job state; gateway owns GPU keys and holds no tenant data at rest                      | Key-custody isolation (the ai-gateway rule applied to GPUs), egress enforcement at the dialer, independent lifecycle for minutes-long jobs |
| D-2 | Dispatch = nest→gateway HTTP + idempotent callback + lease/reconciler; bytes via per-job signed URLs; <b>no new queue infra</b> (Redis for leases only)                                   | Reuses proven patterns; avoids a second durable-queue system before Temporal exists                                                        |
| D-3 | <code>creative_assets</code> is <b>extended in place</b> and upgraded to the third wall; <code>CreativeController</code> moves into the module with the legacy path preserved as an alias | One asset table (no DAM/studio split-brain); shipped studio + trainer contracts unbroken                                                   |
| D-4 | Module key <code>creative</code> , tables <code>creative_*</code> , hr-style controller path + third-wall RLS; <b>sole no-RLS table</b> = <code>creative_workflows</code>                 | Newest-vertical conventions; license source of truth must not fork per tenant                                                              |

| #    | DECISION                                                                                                                                                                                                                                 | WHY                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| D-5  | <b>License wall is structural:</b> workflow manifests → gateway license stamp → asset <code>license_class</code> = weakest of the chain → reuse/deliverable paths refuse chains $\notin$ {commercial, licensed}; no override flag exists | The one legal exposure that can't be walked back after delivery; policy would drift, structure won't                                     |
| D-6  | <b>Two-gate model:</b> spend gate (stop-loss chain + WS4 one-shot approval) + reuse gate (human WS4 approval verifying the license chain)                                                                                                | Generative spend and generative publication are different risks; one gate would either rubber-stamp money or approval-fatigue the studio |
| D-7  | <b>One render ledger platform-wide:</b> sibling-module requests book once with <code>requester_module</code> + client attribution — refines SMM D-9 (flagged \$01, for owner awareness)                                                  | A dollar of GPU must exist exactly once in metering; billing rollups depend on it                                                        |
| D-8  | Renditions = <b>stateless signed imgproxy URLs</b> , no rendition rows; video derivatives are asset versions; preset crops are code-level constants                                                                                      | The whole point of imgproxy is no materialized rendition state; rows would rot                                                           |
| D-9  | CLIP/BLIP/pHash run as <b>analyze render jobs on a CPU worker backend</b> through the same gateway; embeddings are a dual-mode operational column; gateway <code>/embed</code> is never used for asset search                            | Uniform job plumbing, ~\$0 cost, fail-soft ingest; avoids the cross-modal embedding-space mismatch pitfall                               |
| D-10 | Console = Creatives dept, two craft groups: Studio (Image Studio · Generate & Edit · Upscale · Video) + Library (Assets · Brand Kits · Jobs & Usage)                                                                                     | Dept-template doctrine; production vs. supply is the natural split                                                                       |
| D-11 | Video routing per the lock: Wan 2.2 serverless default; commercial API only inside the monthly envelope with per-clip approval above threshold; premium video = scope opt-in like FLUX                                                   | Cost class is 10–50× image; the envelope is the owner's stated budget                                                                    |
| D-12 | Cross-dept consumption = <b>approved-only module endpoints + signed renditions</b> ; consumers never read <code>creative_*</code> tables                                                                                                 | Module boundary enforced by the DB, not convention; rights gating is one choke point                                                     |
| D-13 | Migration block assumption <b>0036+</b> (0034/0035 drawn by search); CR-01 takes next-unused at merge time, coordinating with SMM-01                                                                                                     | Ledger truth as of 2026-07-23; the migrations README's rule handles the race                                                             |

## §15 Glossary

| TERM                    | MEANING                                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Creative Render Gateway | The Go service ( <code>render-gateway-go</code> ) that holds all GPU/model provider keys and executes render jobs; the mirror of ai-gateway-go for diffusion/video workloads                                          |
| RenderBackend           | A capability-based driver interface (serverless / self-host ComfyUI / commercial API) the gateway's router selects between per job, filtered by capability + license class + cost cap + scope + health                |
| License class           | One of <code>commercial</code> , <code>licensed</code> , or <code>noncommercial</code> — computed for an asset as the weakest link of its full derivation chain; drives the router's license floor and the reuse gate |
| Render job              | A <code>creative_render_jobs</code> row — the RLS-scoped, state-machine-authoritative unit of GPU work, from creation through dispatch, execution, and callback                                                       |
| Spend gate              | The pre-dispatch stop-loss chain (scope → license → estimate/threshold → per-cost-class MTD caps → advisory-lock) that refuses a job before GPU money is spent                                                        |
| Reuse gate              | The human WS4 approval that promotes a private asset to cross-department / client-deliverable visibility, verifying every element of its license chain                                                                |
| creative_workflows      | The global, no-RLS, admin-write-only table of versioned ComfyUI workflow JSON + model license manifests — the single source of license truth                                                                          |
| Third-wall RLS          | FORCE-RLS predicate combining tenant scope with per-module enablement, per the hr-migration convention                                                                                                                |
| DAM                     | Digital Asset Management — here, the extended <code>creative_assets</code> domain (versions, collections, brand kits, rights, provenance, CLIP/BLIP search) rather than an adopted external product                   |

| TERM             | MEANING                                                                                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| imgproxy         | The stateless, signed-URL rendition sidecar (MIT, Go/libvips) serving on-the-fly thumbnails/crops straight off storage — no materialized rendition rows                                         |
| Analyze job      | A render job of capability <code>analyze</code> running CLIP ViT-B/32 + BLIP-base on a cheap CPU worker backend, for embeddings/captions/tags — never via the LLM gateway's <code>/embed</code> |
| Requester module | The attribution field on a render job/ledger row identifying which department (creative/social/search/webdesk/agent/automation) asked for the GPU spend, so it books exactly once               |

GAIADA Creative — Engineering Blueprint v1.0

Status: Design blueprint · Expansion Planned · Studio Prototyped (v0.0.0) · 2026-07-23 · Creative Dept · Platform · Classification: Internal

Architecture conforms to the 4 locked foundation decisions (GPU posture, image/video licensing hybrids, build-light DAM). Feeds /army decomposition (§12). Not a commitment of dates. Cross-references: `docs/blueprints/creative-design.md` · `docs/blueprints/creative-foundation.md` · `docs/blueprints/seo-sem-design.md` · `docs/blueprints/smm-design.md` · `docs/modules/MODULES.md` · `docs/FRONTEND-BFF-CONTRACT.md` .